

- 8 (a) Electric force is independent of speed of the charged particle, whereas magnetic force is proportional to speed. [1]

Electric force is directed along E-field direction, whereas magnetic force is directed perpendicular to B-field direction. [1]

(Note: Answers should compare both Electric force and Magnetic force. Elaborating on only 1 force is not sufficient.)

Penalise if student mentioned Electric force is in the direction of E-Field.)

- (b) (i) Since charge is positive, electric force on charge is directed towards –ve plate (downwards). [1]

For the charge to move without deflection, the magnetic force should be directed towards the +ve plate (upwards). [1]

Using Fleming's left hand rule, with magnetic force directed towards +ve plate and conventional current directed towards the right, magnetic field should be directed into the page. (award 1 m for using FLHR correctly) [1]

(ii) $Bqv = q(V/d)$
 $Bv = V/d$ [1]
 $B (1.37 \times 10^6) = 3600 / 0.0275$
 $B = 0.0956 \text{ T}$ [1]

- (iii) Since magnetic force is always directed perpendicular to the velocity of the particle, [1]
the magnetic force provides the centripetal force. [1]
Hence, the particle will move in a circular path.

(iv) Mass of the charged particle = $(6.64 \times 10^{-27}) / (1.66 \times 10^{-27}) = 4u$ [1]
 $Bqv = mv^2 / r$
 $q = mv^2 / Br$
 $= (6.64 \times 10^{-27})(1.37 \times 10^6) / (0.0956)(0.300)$
 $= 3.2 \times 10^{-19} \text{ C}$ [1]

Since the charged particle has mass 4u and charge +2e, it is possibly an alpha particle.

(Note: Give only 1 m if student uses q/m method)

(v) $qE = ma_y$

$$a_y = \frac{(3.2 \times 10^{-19})\left(\frac{3600}{0.0275}\right)}{(6.64 \times 10^{-27})} = 6.309 \times 10^{12} \quad [1]$$

Using $s_y = u_y t + \frac{1}{2} a_y t^2$, we get $\frac{1}{2} (0.0275) = 0 + \frac{1}{2} (6.309 \times 10^{12})(t^2)$
 $t = 6.602 \times 10^{-8} \text{ s} \quad [1]$

Therefore, $s_x = u_x t = (1.37 \times 10^6)(6.602 \times 10^{-8}) = 0.0905 \text{ m} \quad [1]$

Since $s_x < 0.10 \text{ m}$, the charged particle will collide with the plate.

(Note: Penalise 1m if no or wrong conclusion)

ALTERNATIVELY,

$qE = ma_y$

$$a_y = \frac{(3.2 \times 10^{-19})\left(\frac{3600}{0.0275}\right)}{(6.64 \times 10^{-27})} = 6.309 \times 10^{12} \quad [1]$$

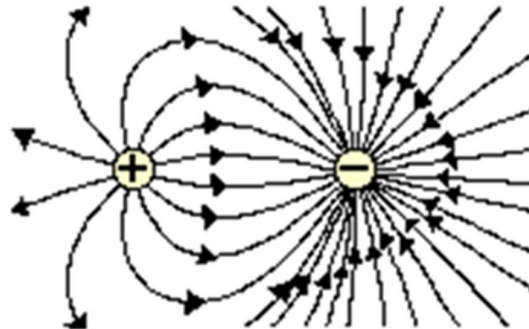
Using $s_y = u_y t + \frac{1}{2} a_y t^2$, we get $\frac{1}{2} (0.0275) = 0 + \frac{1}{2} (6.309 \times 10^{12})(t^2)$
 $t = 6.602 \times 10^{-8} \text{ s} \quad [1]$

Minimum time required to pass the plate horizontally = $0.10 / (1.37 \times 10^6)$
 $= 7.3 \times 10^{-8} \text{ s} \quad [1]$

Since $t < \text{minimum time required}$, the charge particle will collide with the plate.

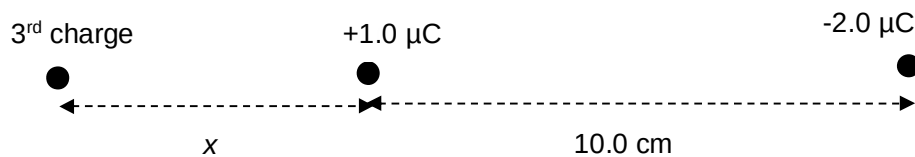
(Note: Penalise 1m if no or wrong conclusion)

(c) (i)



- correct direction of the field lines [1]
- turning point nearer to the (weaker) positive charge and more field lines entering the (stronger) negative charge. [1]

(ii)



Evidence in student's working showing 3rd charge to be on the LEFT of the +1.0 μC charge (either through diagram or indicate qualitatively in final answer) [1]

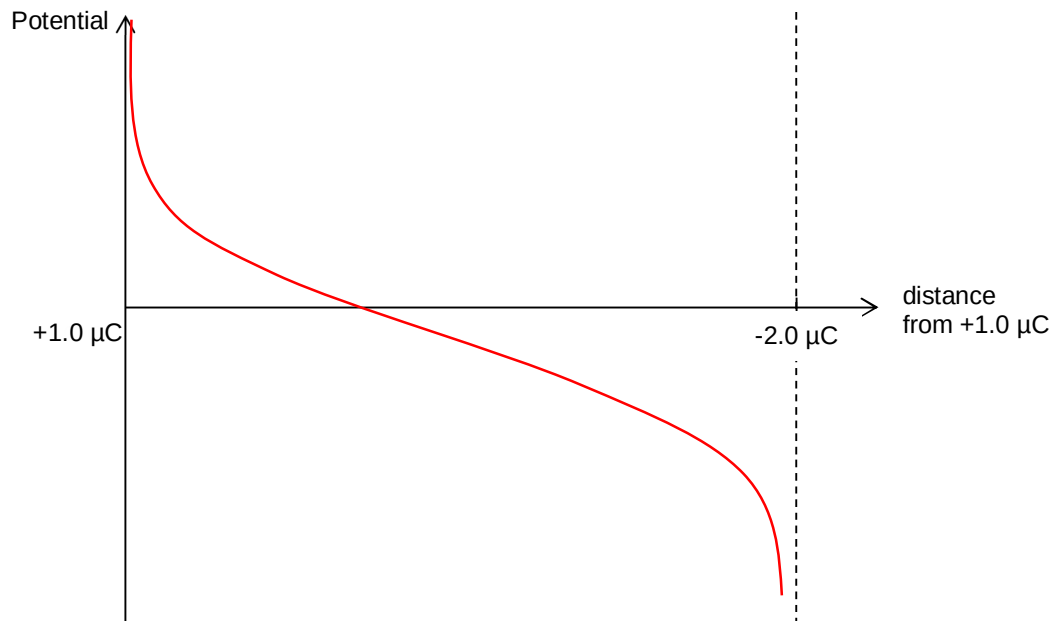
$$\frac{(1.0 \times 10^{-6})q}{4\pi\epsilon_0 x^2} = \frac{(2.0 \times 10^{-6})q}{4\pi\epsilon_0 (0.10 + x)^2}$$

$$0.10 + x = \sqrt{2}x$$

$$x \equiv 0.10 / (\sqrt{2} - 1) \equiv 0.24 \text{ m}$$

[1]

(iii)



- correct shape (asymptotic at both end)
- x-intercept nearer to the +1.0 μC charge

[1]

[1]